



Department of Electrical and Computer
Engineering
Instructor : Hassan Tfaily

Lab 4 : “Interrupt”

Introduction and objectives

Interrupt is a mechanism that forces the CPU to stop the normal program execution to respond to internal (ex: timer overflow)or external (ex: button pressed) events.

It will execute a special function called Interrupt Service Routine ISR ,then returns back to continue executing the remaining instructions of the main program.

In this lab, you will be able to configure , enable and monitor *an external* maskable interrupt IRQ and understand how it influences the program flow .

Pre Lab4 :

In this preliminary task, you are required to answer the following questions :

Ex1:

- HCS12 processors handle maskable and non-maskable interrupts.
Define the 2 types of interrupt and state the differences between them.
- Describe in details how do the processor handle interrupts.
(Program Counter, Stack, Vector Address, ISR ..).
- what is the IRQ interrupt ?
When is it triggered?
To which pin of the microcontroller it is connected ?
- By referring to the “MC9S12XS256RMV1” datasheet (located in folder Microcontroller), explain the role of the following bits of IRQCR (or INTCR) control

register :

- IRQE
- IRQEN

e) Get the Vector Address of the IRQ interrupt from the interrupt vector locations table (refer to “MC9S12XS256RMV1” datasheet).

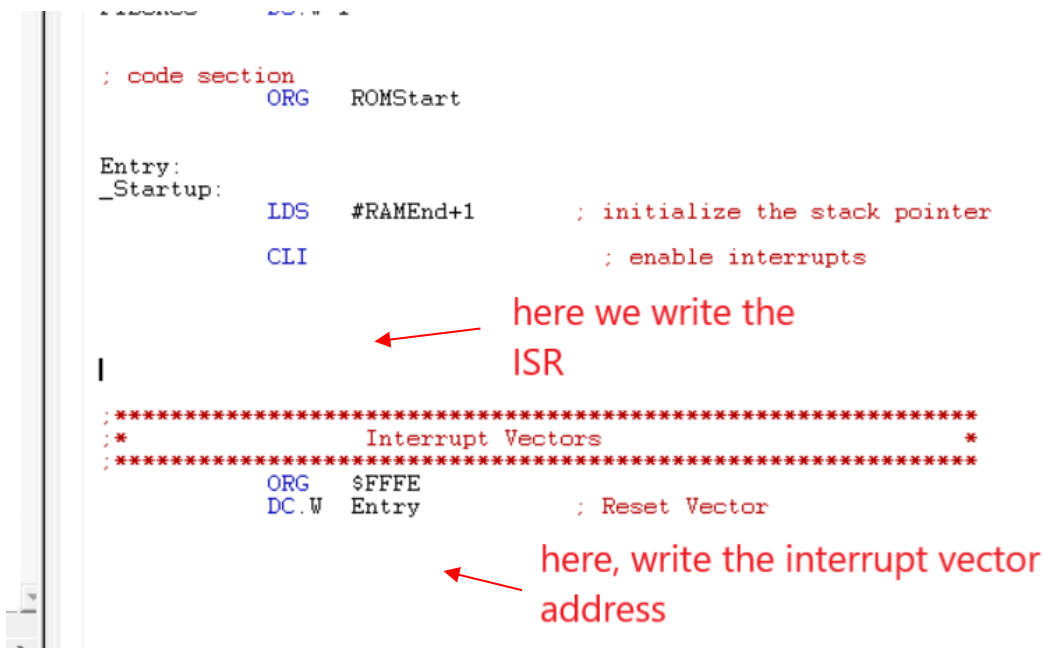
This address is used by the PC to access the Interrupt service Routine ISR whenever the IRQ is triggered .

The ISR is the subroutine that is executed every time the IRQ is generated.

This address will be placed as shown in the figure below:

ORG \$ vector address

DC.W ISRlabel



Lab4: “ Interrupt”

Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the 3 tasks below:

Task1:

The first program should do the following:

Using IRQ , the light must be shifted to the left from LED1 to LED4 EVERY TIME the push button PB1 is pressed .

Task2:

The second program should do the following:

Alternate the LEDs’ shifting either to the right or to the left using IRQ as follows:

Initially the program will shift the light to the left from LED1 to LED4 with a time delay of 200 ms .

Once PB1 is pressed, the LEDs will start shifting to the right with the same time delay.

The next time PB1 is pressed, the LEDs will shift to the left again and the cycle will alternate continuously.

Task3:

The third program should do the following:

Count down from F to 0 displaying the results, in binary, on the LEDs with a delay time of 200ms .

(when the counter is F → LEDs are all ON , counter is 0 → LEDs are all OFF).

Using IRQ , the LEDs will blink ON/OFF for 4 times once PB1 is pressed, then the counter will continue counting down *(from the last value reached).*
